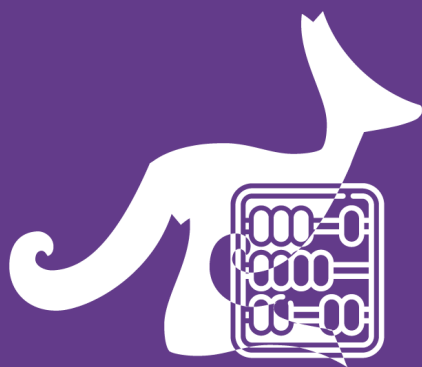




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a

2B pencil.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

Good Luck!

预祝考试顺利

**Year 7
&
Year 8**

3 points

1. Which of the following symbols for signs of the Zodiac has an axis of symmetry?

以下哪一个星座图像有一条对称轴?

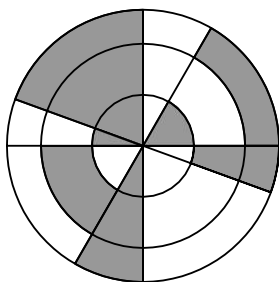
- (A)  Sagittarius 射手座
(B)  Scorpio 天蝎座
(C)  Leo 狮子座
(D)  Cancer 巨蟹座
(E)  Capricorn 摩羯座

$$\frac{20 \times 21}{2 + 0 + 2 + 1}$$

2. The figure shows three concentric circles with four lines passing through their common centre.

What percentage of the figure is shaded?

下图所示有三个同心圆，四条直线均经过其圆心。请问阴影部分面积占整个图形面积的比例是多少?



- (A) 30% (B) 35% (C) 40% (D) 45% (E) 50%

3. What is the value of $\frac{20 \times 21}{2 + 0 + 2 + 1}$?

请问 $\frac{20 \times 21}{2 + 0 + 2 + 1}$ 的值是多少?

- (A) 42 (B) 64 (C) 80 (D) 84 (E) 105

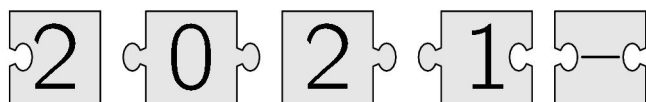
4. How many four-digit numbers have the property that their digits, from left to right, are consecutive and in ascending order?

一个四位数，其各数位上的数字从左到右为连续递增数字。请问这样的四位数有多少个?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

- # 5. When the five pieces shown are fitted together correctly, the result is a rectangle with a calculation written on it. What is the answer to this calculation?

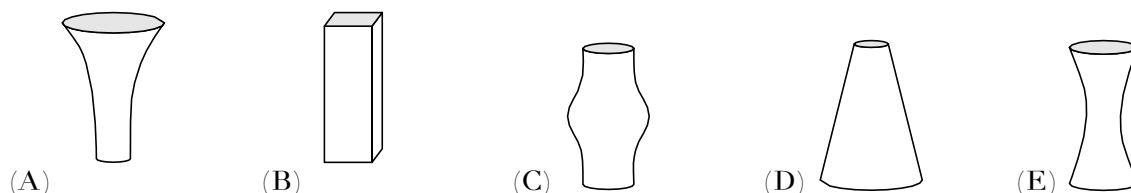
将下列 5 个拼图碎片正确拼接在一起后，会出现写有一道算式题的矩形。请问这个算式的答案是多少？



- (A) -100 (B) -8 (C) -1 (D) 199 (E) 208

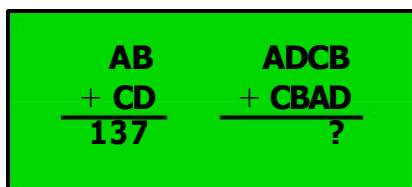
- # 6. Each of the five vases shown has the same height and each has a volume of 1 litre. Half a litre of water is poured into each vase. In which vase would the level of the water be the highest?

下图所示的五个花瓶高度相同，容量均为 1 升。现在往每个花瓶中都倒入半升水，请问哪个花瓶中水的高度最高？



- # 7. A student correctly added the two two-digit numbers on the left of the board and got the answer 137.

一个学生正确计算出下图左侧两个两位数的和为137。

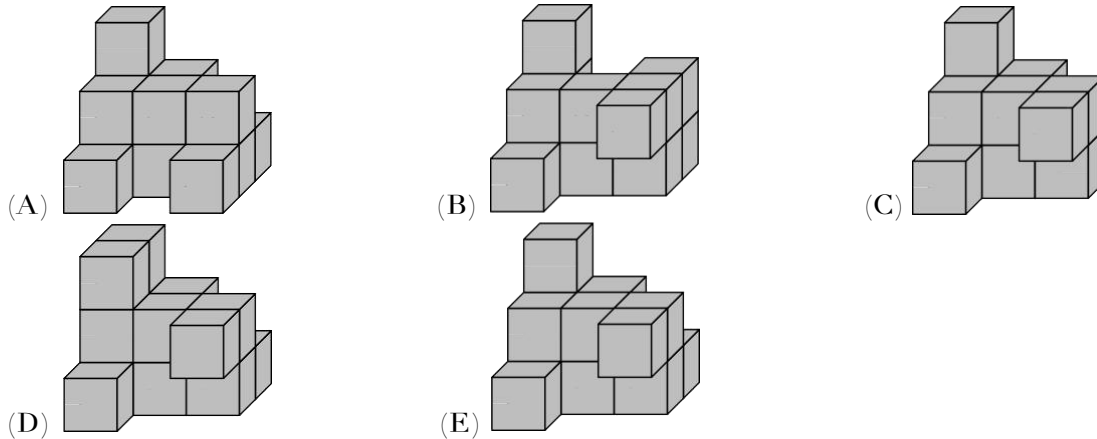
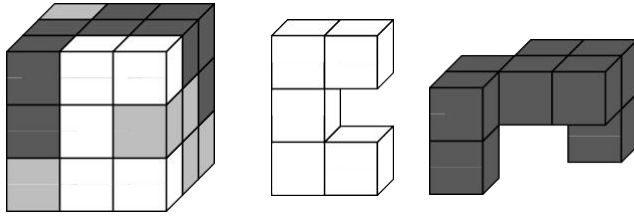


What answer will he get if he adds the two four-digit numbers on the right of the board?
请问图中右侧两个四位数相加之和是多少？

- (A) 13737 (B) 13837 (C) 14747 (D) 23737 (E) 137137

- # 8. A $3 \times 3 \times 3$ cube is made from white, grey and black $1 \times 1 \times 1$ cubes, as shown in the first diagram. The other two diagrams show the white part and the black part of the cube. Which of the following diagrams shows the grey part?

一个 $3 \times 3 \times 3$ 的大立方体是由白色、灰色和黑色的 $1 \times 1 \times 1$ 的小立方体组建而成的，如下左图所示。另外两个图形分别是大立方体中白色和黑色部分。请问以下哪一个是灰色部分的形状？



9. A bike lock has four wheels numbered with the digits 0 to 9 in order. Each of the four wheels is rotated by 180° from the code shown in the first diagram to get the correct code. What is the correct code for the bike lock?

一自行车锁有四个滚轮，每个滚轮上都按顺序标有数字 0 到 9。从下图所示的滚轮位置开始，将每个滚轮按图示方向旋转 180° 后即可转至正确的解锁密码。请问这个自行车锁的密码是多少？



10. Byron is 5cm taller than Aaron, but 10cm shorter than Caron. Darren is 10cm taller than Caron, but 5cm shorter than Erin. Which of the following statements is true?

Byron 比 Aaron 高 5 厘米，但是比 Caron 矮 10 厘米。Darren 比 Caron 高 10 厘米，但是比 Erin 矮 5 厘米。请问以下哪个表述是正确的？

- (A) Aaron and Erin are equal heights Aaron 和 Erin 的身高相同
- (B) Aaron is 10cm taller than Erin Aaron 比 Erin 高 10 厘米
- (C) Aaron is 10cm shorter than Erin Aaron 比 Erin 矮 10 厘米
- (D) Aaron is 30cm taller than Erin Aaron 比 Erin 高 30 厘米
- (E) Aaron is 30cm shorter than Erin Aaron 比 Erin 矮 30 厘米

4 points

- # 11. A rectangular chocolate bar is made of equal squares. Neil breaks off two complete strips of squares and eats the 12 squares he obtains. Later, Jack breaks off one complete strip of squares from the same bar and eats the 9 squares he obtains. How many squares of chocolate are left in the bar?

一块长方形巧克力由相同大小的小巧克力块组成。Neil 掰下两整条巧克力，并吃掉了掰下的这 12 小块巧克力。之后，Jack 又掰下一整条巧克力，并吃掉了掰下的这 9 小块巧克力。请问这个长方形巧克力上还剩多少个小巧克力块？

- (A) 72 (B) 63 (C) 54 (D) 45 (E) 36

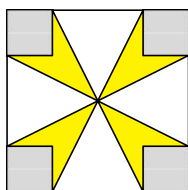
- # 12. A jar one fifth filled with water weighs 560 g. The same jar four fifths filled with water weighs 740g. What is the weight of the empty jar?

一个罐子中如果倒入五分之一的的水，重量为 560 g；如果倒入五分之四的水，重量变为 740 g。请问空罐有多重？

- (A) 60 g (B) 112 g (C) 180 g (D) 300 g (E) 500 g

- # 13. The area of the large square is 16 cm^2 and the area of each small square is 1 cm^2 . What is the total area of the yellow flower?

下图中大正方形的面积是 16 cm^2 ，每个小正方形的面积均为 1 cm^2 。请问图中黄色花形的面积是多少？



- (A) 3 cm^2 (B) $\frac{7}{2} \text{ cm}^2$ (C) 4 cm^2 (D) $\frac{11}{2} \text{ cm}^2$ (E) 6 cm^2

- # 14. Costa is building a new fence in his garden. He uses 25 planks of wood, each of which are 30cm long. He arranges these planks so that there is the same slight overlap between any two adjacent planks.

Costa 正在花园里建造一个新栅栏。他用了 25 块木板，每个木板长 30 厘米。他将这些木板进行排列，使得每两块相邻的木板有同样的部分重叠，如图所示。

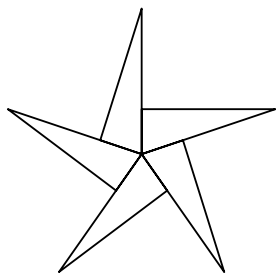


The total length of Costa's new fence is 6.9 metres. What is the length in centimetres of the overlap between any pair of adjacent planks?

已知Costa的新栅栏总长为 6.9 米。请问每两块相邻木板的重叠部分的长度是多少厘米？

- (A) 2.4 (B) 2.5 (C) 3 (D) 4.8 (E) 5

- # 15. Five identical right-angled triangles can be arranged so that their larger acute angles touch to form the star shown in the diagram.



将五个相同的直角三角形中较大的锐角互相拼接，可以形成下图所示的星星形状。

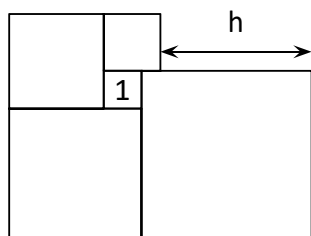
It is also possible to form a different star by arranging more of these triangles so that their smaller acute angles touch. How many triangles are needed to form the second star?

如果把这种直角三角形中较小的锐角互相拼接，也可以形成一个不同的星星形状。请问要形成另外一种星星形状需要多少个三角形？

- (A) 10 (B) 12 (C) 18 (D) 20 (E) 24

- # 16. Five squares are positioned as shown. The small square indicated has area 1. What is the value of h ?

五个正方形的排列位置如图所示。已知其中最小正方形的面积为 1，请问 h 的值是多少？



- (A) 3 (B) 3.5 (C) 4 (D) 4.2 (E) 4.5

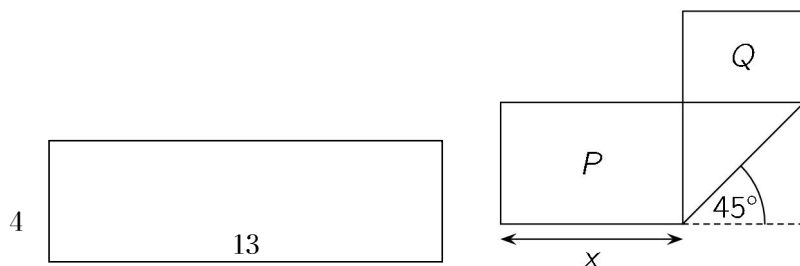
- # 17. There are 20 questions in a quiz. Each correct answer scores 7 points, each wrong answer scores -4 points, and each question left blank scores 0 points. Eric took the quiz and scored 100 points. How many questions did he leave blank?

一场测试中有 20 道题目。每道题目回答正确得 7 分，回答错误扣 4 分，不作答得 0 分。Eric 参加了这场测试，最终得到 100 分。请问他有多少道题目没有作答？

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

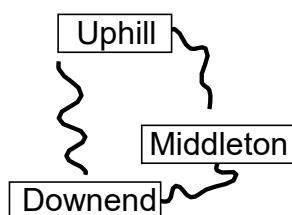
- # 18. A rectangular strip of paper of dimensions 4×13 is folded as shown in the diagram. Two rectangles are formed with areas P and Q where $P = 2Q$. What is the value of x ?

将一张面积为 4×13 的长方形纸条按下图方式折叠，形成两个面积分别为 P 和 Q 的小矩形，其中 $P = 2Q$ 。请问 x 的值是多少？



- (A) 5 (B) 5.5 (C) 6 (D) 6.5 (E) $4\sqrt{2}$
- # 19. A box of fruit contains twice as many apples as pears. Christy and Lily divided them up so that Christy had twice as many pieces of fruit as Lily. Which one of the following statements is always true?
- 一箱水果里面苹果的数量是梨的两倍。Christy和Lily将这些水果分成两份，Christy分得的水果数量是Lily的两倍。请问以下哪个表述永远成立？
- (A) Christy took at least one pear. Christy 至少有 1 个梨
- (B) Christy took twice as many apples as pears. Christy 拥有苹果的数量是梨的2倍
- (C) Christy took twice as many apples as Lily. Christy 拥有苹果的数量是Lily苹果数量的2倍
- (D) Christy took as many apples as Lily got pears. Christy 拥有苹果的数量和Lily拥有梨的数量一样多
- (E) Christy took as many pears as Lily got apples. Christy 拥有梨的数量和Lily拥有苹果的数量一样多
- # 20. Three villages are connected by paths as shown. From Downend to Uphill, the detour via Middleton is 1km longer than the direct path. From Downend to Middleton, the detour via Uphill is 5km longer than the direct path. From Uphill to Middleton, the detour via Downend is 7km longer than the direct path. How long is the shortest of the three direct paths between the villages?

如下图，三个村落由道路相连。从村落 Downend 到村落 Uphill 的两条路线中，经过村落 Middleton 的路线比直达路线多 1 千米；从村落 Downend 到村落 Middleton 的两条路线中，经过村落 Uphill 的路线比直达路线多 5 千米；从村落 Uphill 到村落 Middleton 的两条路线中，经过村落 Downend 的路线比直达路线多 7 千米。请问连接村落的三条道路中，最短的一条道路有多长？



- (A) 1km (B) 2km (C) 3km (D) 4km (E) 5km

5 points

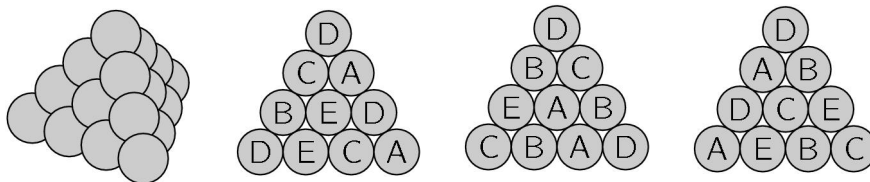
- # 21. In a particular fraction the numerator and denominator are both positive. The numerator of this fraction is increased by 40%. By what percentage should its denominator be decreased so that the new fraction is double the original fraction?

一个分数的分子和分母都是正数。将该分数的分子增加40%后，请问分母应该同步减少多少才能使新得到的分数是原来分数的两倍？

- (A) 10% (B) 20% (C) 30% (D) 40% (E) 50%

- # 22. A triangular pyramid is built with 20 cannon balls, as shown. Each cannon ball is labelled with one of A, B, C, D or E. There are four cannon balls with each type of label. The picture shows the labels on the cannon balls on three of the faces of the pyramid. What is the label on the hidden cannon ball in the middle of the fourth face?

如图所示，20个相同的炮弹搭建成一个三棱锥形状。每个炮弹上标记有一个字母 A、B、C、D 或 E，每个字母均对应 4 个炮弹。下图所示为这个三棱锥三个面上标记的字母。请问第四个面中间位置隐藏的炮弹上标记的字母是什么？



- (A) A (B) B (C) C (D) D (E) E

- # 23. The 6-digit number $2ABCDE$ is multiplied by 3 and the result is the 6-digit number $ABCDE2$. What is the sum of the digits of this number?

一个六位数 $2ABCDE$ 乘以 3 后，得到一个新六位数 $ABCDE2$ 。请问这个数各个数位上的数字相加之和是多少？

- (A) 24 (B) 27 (C) 30 (D) 33 (E) 36

- # 24. A box contains only green, red, blue and yellow counters. There is always at least one green counter amongst any 27 counters chosen from the box; always at least one red counter amongst any 25 counters chosen; always at least one blue amongst any 22 counters chosen and always at least one yellow amongst any 17 counters chosen. What is the largest number of counters that could be in the box?

盒子中装有绿色、红色、蓝色和黄色四种颜色的筹码。从盒子中任意取出 27 个筹码，其中至少有一个是绿色的；任意取出 25 个筹码，其中至少有一个是红色的；任意取出 22 个筹码，其中至少有一个是蓝色的；任意取出 17 个筹码，其中至少有一个是黄色的。请问盒子中最多有多少个筹码？

- (A) 27 (B) 29 (C) 51 (D) 87 (E) 91

- # 25. A soccer ball is made of white hexagons and black pentagons, as seen in the picture. There are a total of 12 pentagons. How many hexagons are there?

如下图，足球由白色的六边形块和黑色的五边形块构成。已知这个足球共有 12 个五边形块，请问这个足球有多少个六边形块？



- (A) 12 (B) 15 (C) 18 (D) 20 (E) 24

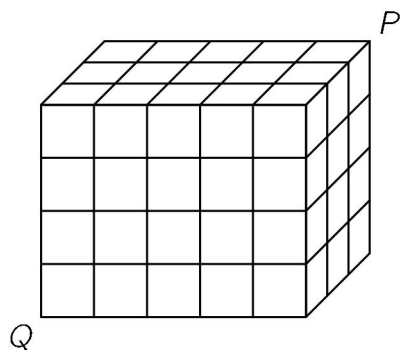
- # 26. 2021 coloured kangaroos are arranged in a row and are numbered from 1 to 2021. Each kangaroo is coloured either red, grey or blue. Amongst any three consecutive kangaroos, there are always kangaroos of all three colours. Bruce guesses the colours of five kangaroos. These are his guesses: Kangaroo 2 is grey; Kangaroo 20 is blue; Kangaroo 202 is red; Kangaroo 1002 is blue; Kangaroo 2021 is grey. Only one of his guesses is wrong. What is the number of the kangaroo whose colour he guessed incorrectly?

将 2021 只彩色袋鼠排成一行，依次编号为 1 到 2021。袋鼠的颜色有红色、灰色、蓝色三种，任意三个相邻的袋鼠，均包含这三种颜色。Bruce 猜测了五只袋鼠的颜色。他的猜测如下：2 号袋鼠是灰色的；20 号袋鼠是蓝色的；202 号袋鼠是红色的；1002 号袋鼠是蓝色的；2021 号袋鼠是灰色的。他的猜测中只有一个是错误的。请问他猜错了几号袋鼠的颜色？

- (A) 2 (B) 20 (C) 202 (D) 1002 (E) 2021

- # 27. A $3 \times 4 \times 5$ cuboid consists of 60 identical small cubes. A termite eats its way along the diagonal from P to Q . This diagonal does not intersect the edges of any small cube inside the cuboid. How many of the small cubes does it pass through on its journey?

一个 $3 \times 4 \times 5$ 的长方体由 60 个完全相同的小立方体组成。一只白蚁沿着对角线从点 P 吃到点 Q ，这条对角线不与长方体内部任何小立方体的边相交。请问白蚁一共经过了多少个小立方体？



- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

- # 28. In a tournament each of the six teams plays one match against every other team. In each round of matches, three take place simultaneously. A TV station has already decided which match it will broadcast for each round, as shown in the diagram. In which round will team D play against team F?

在一场锦标赛中，六支球队每支都要和其他球队进行一场比赛。在每轮比赛中，有三场比赛同时进行。电视台已经确定好每轮转播的比赛场次，安排如下图所示。请问 D 队将会在哪一轮对战 F 队？

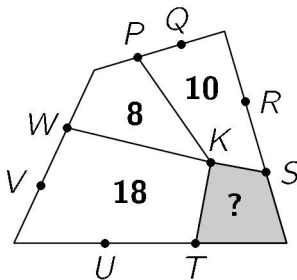
1	2	3	4	5
A-B	C-D	A-E	E-F	A-C

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5
- # 29. In a town there are 21 knights who always tell the truth and 2000 knaves who always lie. A wizard divided 2020 of these 2021 people into 1010 pairs. Every person in a pair described the other person as either a knight or a knave. As a result, 2000 people were called knights and 20 people were called knaves. How many pairs of two knaves were there?
- 在一个小镇里，有 21 个只说真话的骑士和 2000 个只说谎的无赖。一位巫师将这 2021 个人中的 2020 个人两两分组，共分为 1010 组。每个组中的人都称对方是骑士或无赖。最后，有 2000 人被对方称为骑士，有 20 人被对方称为无赖。请问有多少个组中的两个人都是无赖？

- (A) 980 (B) 985 (C) 990 (D) 995 (E) 1000

- # 30. The diagram shows a quadrilateral divided into four smaller quadrilaterals with a common vertex K . The other labelled points divide the sides of the large quadrilateral into three equal parts. The numbers indicate the areas of the corresponding small quadrilaterals. What is the area of the shaded quadrilateral?

下图中，一个大四边形被分成四个小四边形，四个小四边形有一个共同的顶点 K 。其他标记的点为大四边形各条边上的三等分点，数字表示对应的小四边形的面积。请问阴影四边形的面积是多少？

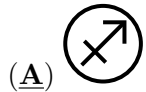


- (A) 4 (B) 5 (C) 6 (D) 6.5 (E) 7

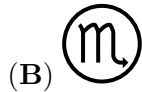
PROBLEMS WITH SOLUTIONS

3 points

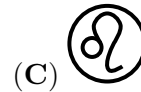
1. Which of the following symbols for signs of the Zodiac has an axis of symmetry?



Sagittarius



Scorpio



Leo



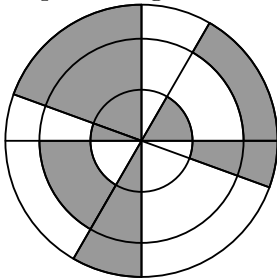
Cancer



Capricorn

SOLUTION:

2. The figure shows three concentric circles with four lines passing through their common centre. What percentage of the figure is shaded?



(A) 30%

(B) 35%

(C) 40%

(D) 45%

(E) 50%

SOLUTION: For every grey area its symmetric is white and vice versa. So grey = white.

3. What is the value of $\frac{20 \cdot 21}{2 + 0 + 2 + 1}$?

(A) 42

(B) 64

(C) 80

(D) 84

(E) 105

SOLUTION: $\frac{20 \cdot 21}{2 + 0 + 2 + 1} = \frac{20 \cdot 21}{5} = 4 \cdot 21 = 84$

4. How many four-digit numbers have the property that their digits, from left to right, are consecutive and in ascending order?

(A) 5

(B) 6

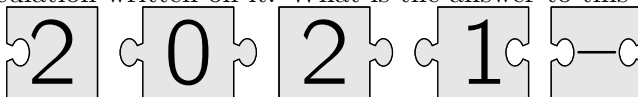
(C) 7

(D) 8

(E) 9

SOLUTION:

5. When the five pieces shown are fitted together correctly, the result is a rectangle with a calculation written on it. What is the answer to this calculation?



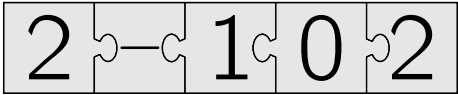
(A) -100

(B) -8

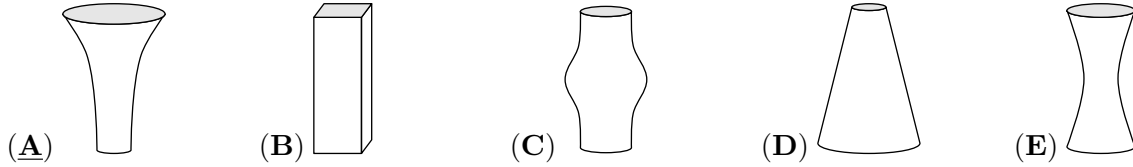
(C) -1

(D) 199

(E) 208

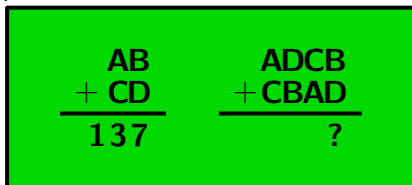
SOLUTION: 

6. Each of the five vases shown has the same height and each has a volume of 1 litre. Half a litre of water is poured into each vase. In which vase would the level of the water be the highest?



SOLUTION:

7. A student correctly added the two two-digit numbers on the left of the board and got the answer 137.

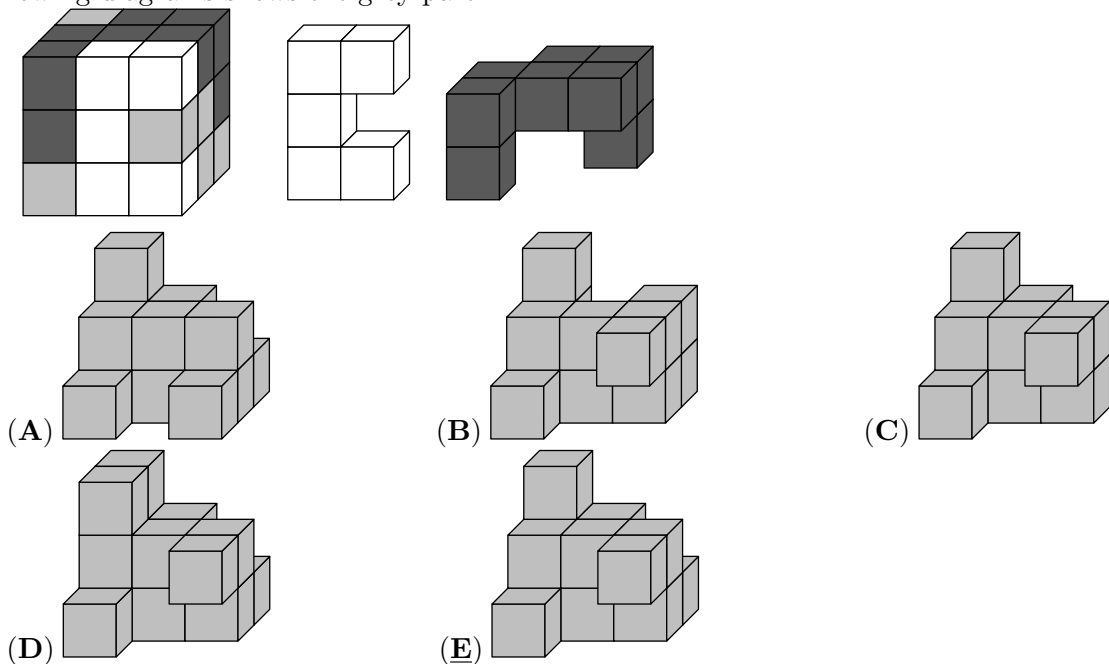


What answer will he get if he adds the two four-digit numbers on the right of the board?

- (A) 13737 (B) 13837 (C) 14747 (D) 23737 (E) 137137

SOLUTION: $\overline{ADCB} + \overline{CBAD} = 10^2(\overline{AD} + \overline{CB}) + (\overline{CB} + \overline{AD}) = 10^2(\overline{AB} + \overline{CD}) + (\overline{AB} + \overline{CD}) = 10^2 \times 137 + 137 = 13700 + 137 = 13837$

8. A $3 \times 3 \times 3$ cube is made from white, grey and black $1 \times 1 \times 1$ cubes, as shown in the first diagram. The other two diagrams show the white part and the black part of the cube. Which of the following diagrams shows the grey part?



SOLUTION:

9. A bike lock has four wheels numbered with the digits 0 to 9 in order. Each of the four wheels is rotated by 180° from the code shown in the first diagram to get the correct code. What is the correct code for the bike lock?



- (A) (B) (C) (D) (E)

SOLUTION:

10. Byron is 5cm taller than Aaron, but 10cm shorter than Caron. Darren is 10cm taller than Caron, but 5cm shorter than Erin. Which of the following statements is true?

- (A) Aaron and Erin are equal heights (B) Aaron is 10cm taller than Erin
 (C) Aaron is 10cm shorter than Erin (D) Aaron is 30cm taller than Erin
 (E) Aaron is 30cm shorter than Erin

SOLUTION: $A < B < C < D < E$ with total difference $5 + 10 + 10 + 5 = 30$ cm.

4 points

11. A rectangular chocolate bar is made of equal squares. Neil breaks off two complete strips of squares and eats the 12 squares he obtains. Later, Jack breaks off one complete strip of squares from the same bar and eats the 9 squares he obtains. How many squares of chocolate are left in the bar?

- (A) 72 (B) 63 (C) 54 (D) 45 (E) 36

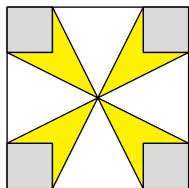
SOLUTION:

12. A jar one fifth filled with water weighs 560 g. The same jar four fifths filled with water weighs 740 g. What is the weight of the empty jar?

- (A) 60 g (B) 112 g (C) 180 g (D) 300 g (E) 500 g

SOLUTION:

13. The area of the large square is 16 cm^2 and the area of each small square is 1 cm^2 . What is the total area of the yellow flower?



- (A) 3 cm^2 (B) $\frac{7}{2} \text{ cm}^2$ (C) 4 cm^2 (D) $\frac{11}{2} \text{ cm}^2$ (E) 6 cm^2

SOLUTION:

14. Costa is building a new fence in his garden. He uses 25 planks of wood, each of which are 30 cm long. He arranges these planks so that there is the same slight overlap between any two adjacent planks.

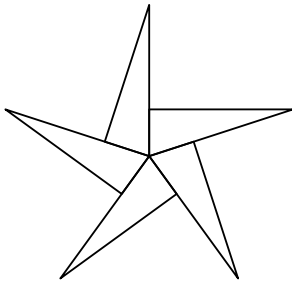


The total length of Costa's new fence is 6.9 metres. What is the length in centimetres of the overlap between any pair of adjacent planks?

- (A) 2.4 (B) 2.5 (C) 3 (D) 4.8 (E) 5

SOLUTION: $25 \times 30 = 750$ so overlap is 60cm. There are 24 overlaps so each is 2.5 cm.

15. Five identical right-angled triangles can be arranged so that their larger acute angles touch to form the star shown in the diagram.

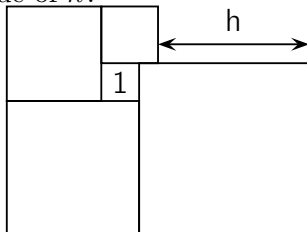


It is also possible to form a different star by arranging more of these triangles so that their smaller acute angles touch. How many triangles are needed to form the second star?

- (A) 10 (B) 12 (C) 18 (D) 20 (E) 24

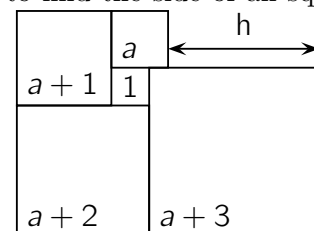
SOLUTION:

16. Five squares are positioned as shown. The small square indicated has area 1. What is the value of h ?



- (A) 3 (B) 3.5 (C) 4 (D) 4.2 (E) 4.5

SOLUTION: Call a the side of the top right square. It is now easy to find the side of all squares in



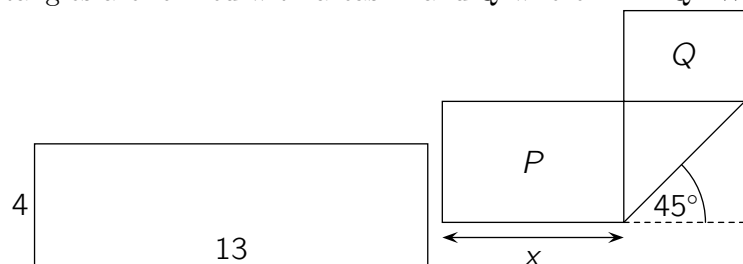
terms of a . Observe $h = (a+2) + (a+3) - a - (a+1) = 4$ (independent of a !)

17. There are 20 questions in a quiz. Each correct answer scores 7 points, each wrong answer scores -4 points, and each question left blank scores 0 points. Eric took the quiz and scored 100 points. How many questions did he leave blank?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

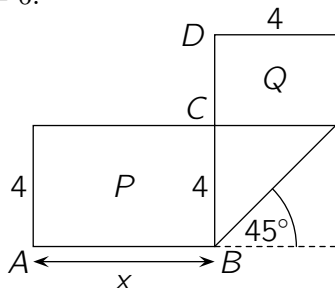
SOLUTION: Firstly, as the value of each blank answer or wrong answer 0 and -4 and the final score 100 are all multiples of 4 while the value of each correct answer 7 is relatively prime to 4, the numbers of correct answers should be divisible by 4 otherwise the overall score will not be a multiple of 4. There are 20 problems, if he gets all correct answers, the score is 140 rather than 100, the next possible number of correct answers that is divisible by 4 will be 16, he gets $16 \times 7 = 112$ points for correct answers, as the final score is 100, there will be $(112 - 100) \div 4 = 3$ wrong answers and $20 - 16 - 3 = 1$ blank answer.// Another solution: Let W be the wrong answers and X the unanswered. Note if any of the wrong questions was answered correctly it would mean $11 - (-4) = 11$ extra points. Also if any of the unanswered questions was answered correctly, it would mean 7 extra points. So the extra $20 \times 7 - 100 = 40$ points he would get by answering everything correctly satisfies $40 = 11W + 7X$. Observe now it follows that W is at most 3. Testing the small values 0, 1, 2, 3 we find the unique solution $W = 3$, $X = 1$ etc.// Another solution: Let W be the wrong answers and X the unanswered. The number of correct answers is $20 - W - X$. So in terms of points we have $7(20 - W - X) - 4W = 100$, which simplifies to $40 = 11W + 7X$, as before.

18. A rectangular strip of paper of dimensions 4×13 is folded as shown in the diagram. Two rectangles are formed with areas P and Q where $P = 2Q$. What is the value of x ?



- (A) 5 (B) 5.5 (C) 6 (D) 6.5 (E) $4\sqrt{2}$

SOLUTION: We have $AB + BD =$ the original side length 13. Because the folding is at 45° , it follows that BC is perpendicular to AB and so equal and parallel to the side 4. So $x + 4 + CD = 13$, thus $CD = 9 - x$ and area $Q = 4(9 - x)$. The equation $P = 2Q$ becomes $4x = 2 \cdot 4(9 - x)$, from where $x = 6$.



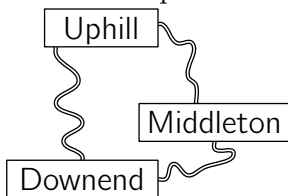
19. A box of fruit contains twice as many apples as pears. Christy and Lily divided them up so that Christy had twice as many pieces of fruit as Lily. Which one of the following statements is always true?

- (A) Christy took at least one pear.

- (B) Christy took twice as many apples as pears.
 (C) Christy took twice as many apples as Lily.
 (D) Christy took as many apples as Lily got pears.
 (E) Christy took as many pears as Lily got apples.

SOLUTION: Since the apples are $\frac{2}{3}$ of the fruits and Christy got $\frac{2}{3}$ of the fruits, one way as dividing the fruits among the two is by giving to Christy all the apples and to Lily all the pears. But this is just one way of dividing the fruits. If Christy wants some pears, she can exchange some of her apples with pears. The question is how many. Well, if she wants to retain the number of fruits she has, it means that for every apple she gives to Lily she must get one pear. This shows that answer (E) is correct.

20. Three villages are connected by paths as shown. From Downend to Uphill, the detour via Middleton is 1km longer than the direct path. From Downend to Middleton, the detour via Uphill is 5km longer than the direct path. From Uphill to Middleton, the detour via Downend is 7km longer than the direct path. How long is the shortest of the three direct paths between the villages?



- (A) 1km (B) 2km (C) 3km (D) 4km (E) 5km

SOLUTION: Just for double checking, solving the obvious linear system for the in between distances among the villages, we will find D to U is 4km, U to M is 3 km, D to M is 6 km

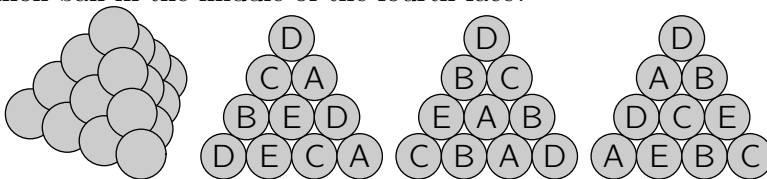
5 points

21. In a particular fraction the numerator and denominator are both positive. The numerator of this fraction is increased by 40%. By what percentage should its denominator be decreased so that the new fraction is double the original fraction?

- (A) 10% (B) 20% (C) 30% (D) 40% (E) 50%

SOLUTION: $1,4/x=2$ that is $x=0,7$ so the denominator should decrease with 30%.

22. A triangular pyramid is built with 20 cannon balls, as shown. Each cannon ball is labelled with one of A, B, C, D or E. There are four cannon balls with each type of label. The picture shows the labels on the cannon balls on three of the faces of the pyramid. What is the label on the hidden cannon ball in the middle of the fourth face?



- (A) A (B) B (C) C (D) D (E) E

SOLUTION: Note that all cannon balls on the edges of the pyramid (as for example those marked yellow) appear on several views. Taking this into account and making sure we do not double count the cannon balls we will find that there are 4 with A, B, C and E, and only 3 with D. It follows that the hidden cannon ball has D on it.

23. The 6-digit number $2ABCDE$ is multiplied by 3 and the result is the 6-digit number $ABCDE2$. What is the sum of the digits of this number?

- (A) 24 (B) 27 (C) 30 (D) 33 (E) 36

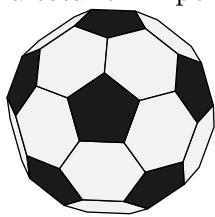
SOLUTION: Denoting $ABCDE$ by x , the condition becomes $3(2 \cdot 10^5 + x) = 10x + 2$. Solving, $x = 85714$ so the number is 285714.

24. A box contains only green, red, blue and yellow counters. There is always at least one green counter amongst any 27 counters chosen from the box; always at least one red counter amongst any 25 counters chosen; always at least one blue amongst any 22 counters chosen and always at least one yellow amongst any 17 counters chosen. What is the largest number of counters that could be in the box?

- (A) 27 (B) 29 (C) 51 (D) 87 (E) 91

SOLUTION: Conditions give $R+B+Y \leq 26, G+B+Y \leq 24, G+R+Y \leq 21, G+R+B \leq 16$. Adding $3(R+B+G+Y) \leq 87$ or $R+B+G+Y \leq 29$. Moreover we can get 29 using $B = 8, G = 3, R = 5, Y = 13$

25. A soccer ball is made of white hexagons and black pentagons, as seen in the picture. There are a total of 12 pentagons. How many hexagons are there?



- (A) 12 (B) 15 (C) 18 (D) 20 (E) 24

SOLUTION:

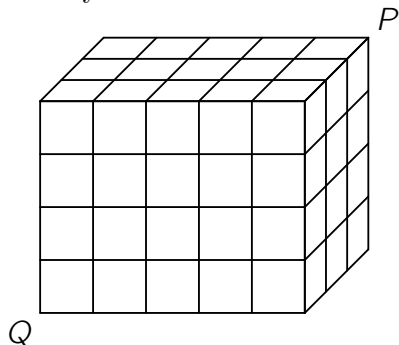
26. 2021 coloured kangaroos are arranged in a row and are numbered from 1 to 2021. Each kangaroo is coloured either red, grey or blue. Amongst any three consecutive kangaroos, there are always kangaroos of all three colours. Bruce guesses the colours of five kangaroos. These are his guesses: Kangaroo 2 is grey; Kangaroo 20 is blue; Kangaroo 202 is red; Kangaroo 1002 is blue; Kangaroo 2021 is grey. Only one of his guesses is wrong. What is the number of the kangaroo whose colour he guessed incorrectly?

- (A) 2 (B) 20 (C) 202 (D) 1002 (E) 2021

SOLUTION: Consider any four consecutive kangaroos. The first three kangaroos have a different colour, as well as the last three kangaroos. Thus, the first and the fourth kangaroos have the same color. Hence, the color of any kangaroo is repeated after three kangaroos. Thus, the kangaroos nr. 2,

nr. 20 and nr. 2021 must have the same colour (since $20 = 3 \times 6 + 2$, $2021 = 3 \times 673 + 2$). Since only one answer is wrong, it is about kangaroo nr. 20.

27. A $3 \times 4 \times 5$ cuboid consists of 60 identical small cubes. A termite eats its way along the diagonal from P to Q . This diagonal does not intersect the edges of any small cube inside the cuboid. How many of the small cubes does it pass through on its journey?



- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

SOLUTION: Each time the termite eats itself through a wall, it enters a new cube. In each direction the termite eats through $n - 1$ walls. Therefore the termite passes through $1 + (3 - 1) + (4 - 1) + (5 - 1) = 10$ little cubes.

28. In a town there are 21 knights who always tell the truth and 2000 knaves who always lie. A wizard divided 2020 of these 2021 people into 1010 pairs. Every person in a pair described the other person as either a knight or a knave. As a result, 2000 people were called knights and 20 people were called knaves. How many pairs of two knaves were there?

- (A) 980 (B) 985 (C) 990 (D) 995 (E) 1000

SOLUTION:

29. In a tournament each of the six teams plays one match against every other team. In each round of matches, three take place simultaneously. A TV station has already decided which match it will broadcast for each round, as shown in the diagram. In which round will team D play against team F?

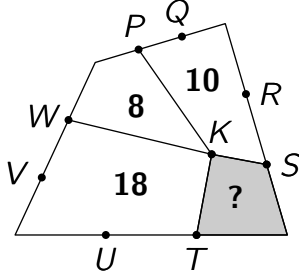
1	2	3	4	5
A-B	C-D	A-E	E-F	A-C

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

1	2	3	4	5
A-B	C-D	A-E	E-F	A-C
C-E	A-F	B-D	A-D	B-F
D-F	B-E	C-F	B-C	D-E

SOLUTION:

30. The diagram shows a quadrilateral divided into four smaller quadrilaterals with a common vertex K . The other labelled points divide the sides of the large quadrilateral into three equal parts. The numbers indicate the areas of the corresponding small quadrilaterals. What is the area of the shaded quadrilateral?



(A) 4

(B) 5

(C) 6

(D) 6.5

(E) 7

SOLUTION: Join K with the vertices. As triangles AKP , PKB have the same altitude and their bases satisfy $PB = 2AP$, it follows that $KPB = 2AKP$. We get similar relationship for all the other triangles. From this it is easy to see that the sum of the two white areas is double the sum of the two grey ones. So $10 + 18 = 2(8 + KSCT)$, so $KSCT = 6$.

